

Background — Draft

IWAI 2026: Paradigm Dynamics in Multi-Agent Motivated Belief

Hallgren & Hyland

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1 Theory-Laden Perception

Kuhn’s three faces. The central phenomenological commitment of this paper is the Kuhnian observation that the paradigm a scientist holds shapes what she is even capable of perceiving (Kuhn, 1962). This is not a methodological lapse to be corrected by more careful experimental design but a structural feature of scientific cognition, and it has three interconnected faces that we will treat as expressions of a single underlying phenomenon. The first face is what a scientist sees in incoming data: the same observation is read differently from inside different paradigms, with the assignment of *anomaly* versus *nuisance* itself being paradigm-dependent (Brewer and Loschky, 2010; Shimony, 1976). The second face is what experiments she chooses to run: the paradigm shapes the space of investigations she finds worth undertaking, which in turn shapes what evidence the community as a whole produces (Halloun, 2016). The third face is what theoretical alternatives she even considers: rival frameworks are not so much rejected as never seriously entertained, with the candidate set of live revisions itself being structured by the prevailing framework (Carrier, 2002; Malone, 1993).

Theory-ladenness as structural feature, not bias. The contemporary literature has substantially clarified what was, in Kuhn’s original framing, sometimes left ambiguous. Siegel (1987) and Malone (1993) reconstruct incommensurability as a condition on referential and inferential overlap between frameworks rather than as a relativist denial of cross-paradigm rationality, and Carrier (2002) demonstrates on the phlogiston case that *empirical comparability* between rival paradigms is possible without their being intertranslatable term-by-term. Cognitive psychology has independently corroborated the perceptual layer of the claim: Brewer and Loschky (2010) review top-down and bottom-up contributions to scientific observation and find that the prior model substantively conditions what counts as a datum, in line with the phenomenology Kuhn was describing. The resulting picture is deflationary in one direction — theory-ladenness is not the collapse of objectivity — and inflationary in another: the influence of the prevailing framework reaches further into the cognitive pipeline than an additive bias term would suggest, touching attention, salience, the felt space of alternatives, and the allocation of experimental effort.

The Chemical Revolution as a worked case. The Chemical Revolution is the canonical historical demonstration of all three faces operating jointly. Chang (2012) argues that phlogiston

theory was not a failed predecessor refuted by Lavoisier’s data but a research programme with its own coherent set of observables, anomalies, and live revisions; its replacement was less a victory of evidence over confusion than a reallocation of what counted as a phenomenon to be explained. Lewowicz (2011) and Boantz (2008) reinforce this reading from inside the chemistry: the phlogiston framework had resources — including a treatment of heat (Boantz, 2008) — whose referents were not so much falsified as relocated under the oxygen framework, exactly the kind of move that the third Kuhnian face predicts. This historical record matters for our model because it shows that the three faces are not independent variables to be added: shifts in what is observed, what is investigated, and what is entertained co-vary, and they co-vary on the timescale of community-wide reorganisation rather than individual reasoning.

Methodological commitment. We take this as the empirical commitment that licenses the model’s complexity. An accumulated-checklist alternative — enumerate biases in perception, biases in experiment choice, and biases in hypothesis generation, then sum them — would be both more defensive and less informative, because it would treat the three faces as separable failure modes rather than as expressions of a single structural fact about how a community-held framework conditions cognition. The stronger rhetorical position, and the one this paper takes, is that a single underlying mechanism is responsible for all three faces, and that the model must therefore couple perception, action, and hypothesis-space in a way that an additive bias account cannot. The remainder of the paper develops that mechanism. In our setting, the shared framework is encoded in the joint structure of agents’ generative models and the channel of inter-agent precision that binds them; the three faces will reappear, respectively, as the likelihood-weighting in inference, as the experiment-selection policy, and as the support of the prior over theories. The case for treating them together, rather than enumerating them separately, rests on the phenomenological commitment defended here.

2 Motivated Belief at the Single-Agent Level

Empirical motivated reasoning. The cognitive-psychological case that belief revision is shaped by goals other than accuracy is long-standing and converged. The canonical statement is Kunda’s argument that reasoning is steered by *directional* as well as accuracy goals: subjects construct the sequence of inferential steps that licenses the conclusion they would prefer to reach, subject to the constraint that the construction remain plausible to themselves (?). This framing absorbed two earlier empirical findings as special cases: the biased-assimilation experiments of ?, in which the same mixed evidence about a contested policy strengthened the prior commitments of partisans on *both* sides; and confirmation-bias phenomena more broadly catalogued by ?. The empirical literature has since accumulated in two directions. The first is Kahan’s cultural-cognition programme, in which the directional goal is identified specifically as *identity protection*: subjects process risk-relevant evidence in whatever way preserves their standing in the affinity groups they belong to (Kahan et al., 2007; Kahan, 2017). The signature empirical finding is that the gap between cultural sub-populations *widens* with numerical and scientific sophistication rather than narrowing (Kahan et al., 2017); Kahan reads this as evidence that the more capable reasoners are more effective at the selective evaluation of identity-incongruent evidence, a behaviour he characterises as *expressive rationality* — rational relative to the agent’s practical stakes in group membership, irrational relative to truth (Kahan, 2015). The mechanism remains contested: Glüer-Pagin and Spectre (2024) argue that what Kahan’s experiments isolate is motivation *to reason* rather than motivated reasoning in the strict sense, and the broader political-science literature now distinguishes carefully between directional and accuracy regimes and the conditions under which one or the other dominates (Bolsen and Palm, 2019; Little et al., 2020; Savolainen, 2022). For our purposes the contested points do not matter: the robust observation is that human belief revision exhibits a systematic, content-sensitive asymmetry in its receptivity to

evidence, and that this asymmetry is not adequately captured by treating the agent as an unbiased Bayesian updater.

Variational formalisation. The active-inference community has begun to give this asymmetry a principled mathematical form. The starting point is the variational free-energy functional, in which an agent’s update from prior to posterior incurs a complexity cost proportional to the Kullback–Leibler divergence between the candidate posterior and the prior (Costa et al., 2020). Hyland and Albarracin (2025) re-foreground this complexity term as a *motivational* quantity: the cost of changing one’s mind is the KL divergence from the current belief to the candidate revision, weighted by an affective utility that reflects how much the agent cares about retaining the prior. Because that divergence is asymmetric, belief revision is path-dependent: evidence that would update a neutral prior into a particular posterior need not, played in reverse, undo a revision already made. This is the formal counterpart of the biased-assimilation and identity-protection findings: preferred beliefs are not merely held more confidently, they are protected by a mathematically explicit revision cost that contrary evidence must overcome. Kruglanski (2020) reaches a sympathetic conclusion from the social-psychological side, arguing that there is no sharp boundary between motivated and non-motivated cognition because *all* inference is directional in the sense that it is steered by what the agent values; the variational reading makes this claim precise by locating the directional weight in the affective utility that multiplies the complexity term. Adjacent strands of the active-inference literature provide the surrounding machinery on which the multi-agent extension will draw, including federated inference under shared world models (Friston et al., 2023) and the free-energy account of belief-revision dynamics more generally (Hopkins, 2016).

Open question. The single-agent picture is now reasonably well developed, both empirically and formally. What it does not yet capture is the phenomenon Kuhn observed at the community scale: that scientific sub-fields exhibit collective motivated belief, sustained across generations, with the directional goal residing not in any individual ego but in the institutional and social structures that bind the community together. The active-inference treatment so far has been single-agent — the cost of mind-change is borne by one mind, against its own prior. The empirical motivated-reasoning literature is in practice multi-agent — the identity being protected is a group identity — but its formalisation has remained at the level of within-individual mechanism. The question this paper takes up is how the variational cost of belief revision generalises to a population of coupled inferential agents in a way that recovers what Kuhn observed: paradigm-level persistence that is sociologically generated, asymmetric in time, and not reducible to the directional preferences of any single agent.

3 From Single-Agent to Multi-Agent: The Scalar Treatment

Scalar Bayesian updating on a graph. The scalar multi-agent setting we adopt sits inside a well-developed tradition that treats opinion change as a linear, convex-combination update on a weighted directed graph. The canonical primitive is the DeGroot averaging model (Boudourides, 2026), in which each agent’s opinion at the next step is a row-stochastic mixture of its neighbours’ current opinions; under mild connectivity assumptions this contracts to consensus. The Friedkin–Johnsen extension breaks that asymptotic consensus by allowing each agent to retain weight on its initial belief through a susceptibility coefficient $s_i \in [0, 1]$, so that the long-run opinion vector is an affine-resolvent functional of the boundary of stubborn agents (Boudourides, 2026). The conservatism parameter in our setup plays the same formal role as the Friedkin–Johnsen susceptibility, read off the variational free-energy decomposition rather than postulated: the cost-of-mind-change term in the tempered-Bayesian update of Hyland and Albarracin (2025) is precisely an asymmetric KL penalty on revising one’s posterior, and at the linear limit its scalar

weight maps to $1 - s_i$. The literature has worked out a great deal of structural detail on this family: closed-form steady states via resolvent / Green-function representations, quantitative convergence rates controlled by the spectral radius of the susceptibility-weighted influence matrix, and explicit sensitivity of the steady state to perturbations of either the influence matrix or the susceptibility profile (Boudourides, 2026). Signed generalisations recover polarisation as a stable attractor when the underlying graph admits a structural-balance decomposition into hostile camps (Proskurnikov et al., 2015), and community-detection methods exploit the same consensus dynamics to read off hierarchical group structure from opinion trajectories (Ren et al., 2020). We inherit this machinery and re-read it in active-inference terms: the influence matrix is a learned channel precision rather than a fixed social-tie weight, and the susceptibility / conservatism is a free-energy quantity rather than a phenomenological parameter.

Multi-armed bandit framing of theory choice. The complementary tradition treats the agent’s problem not as averaging an opinion but as choosing which arm of a multi-armed bandit to pull next given its neighbours’ payoffs, which we draw on explicitly when the scalar opinion is read as a paradigm *choice* rather than a continuous belief. Bala and Goyal (1998) introduce the canonical model: each agent repeatedly chooses among a finite set of risky actions, observes its own and its neighbours’ payoffs, and updates a Bayesian posterior over which action is best. They prove that on connected networks the population converges to consensus on a single action almost surely, but the consensus action need not be the optimal one — early random fluctuations get locked in by social imitation, generating a long tail of pathological lock-ins on inferior arms. Their follow-up (Bala and Goyal, 2001) sharpens this into a quantitative tradeoff between conformism and diversity that prefigures the network-epistemology framing later picked up by ?, whose paradoxical-stubbornness and communication-structure results we treat as derived consequences of the same primitives and develop in Section 6. The Bala–Goyal lineage is the right reference frame for our §3 simulations because it is the minimal model in which (i) the agent has a well-defined Bayesian posterior over a finite paradigm set, (ii) communication enters through observed payoffs rather than through opinion averaging, and (iii) the timescale-separation between learning and convergence generates path-dependent population outcomes — the three features the scalar paradigm experiments are designed to exhibit.

Active-inference coupling. Bringing the two strands together in an active-inference register is the contribution of Albarracin et al. (2022), who couple variational-Bayesian agents through shared observations and characterise echo-chamber bifurcation as a function of the inter-agent precision. Their construction is the immediate ancestor of our §3 setup, with two refinements. First, the inter-agent precision is itself learned, not fixed, so the trust matrix is the only Bayesian-learned relational object in the model and the “susceptibility profile” of the Friedkin–Johnsen analysis becomes endogenous to the dynamics. Second, the tempered-Bayesian update of Hyland and Albarracin (2025) contributes an asymmetric cost-of-mind-change term that does not appear in the symmetric linear-averaging tradition but that is necessary to recover Kuhnian path-dependence at the population level: even a homogeneous network with isotropic trust will, with non-zero conservatism, lock onto its initial-condition basin in a way that pure DeGroot or Bala–Goyal dynamics cannot. The scalar experiments that follow probe how this conservatism parameter interacts with the trust-precision topology and with the affective utility on paradigm values to produce the population behaviours we will then re-read, in §4, as the structural limits of a scalar treatment.

4 What the Scalar Treatment Cannot Capture

What the scalar treatment recovers. Before locating its limits, it is worth being explicit about what the scalar treatment of paradigm commitment already captures. A temperature-

modulated update on a single belief variable produces an asymmetric assimilation: the same datum is integrated by an agent who sits near one attractor with a different effective Bayes factor than by an agent who sits near another, so that observers in different paradigms reach different posteriors from a shared observation (Kuhn, 1962). This is not a degenerate re-description of confirmation bias; it is a principled consequence of the cost of mind-change, and it does some of the work that Kuhn’s first face of theory-ladenness demands. To that extent, the scalar treatment is doing real explanatory labour, and we resist the temptation to throw it away in order to motivate what follows. The point of this section is to mark where its explanatory reach ends.

Structural reorganisation: the Lavoisian case. The clearest pressure on a scalar treatment comes from the Chemical Revolution. Chang (2012) argues that the disagreement between Lavoisier and the late phlogistonians cannot be cashed out as a disagreement over the probability assigned to a shared proposition. What counted as combustion, what counted as a substance, what counted as the conservation of something, and what an experimental yield was even evidence *for*, all shifted together; the very referents of the disputants’ terms drift across the transition (Lewowicz, 2011). Carrier (2002) makes the related point that intertheoretic comparison across the phlogiston/ oxygen divide requires reconstructing each side’s network of inferential commitments rather than aligning a common variable; the incommensurability at stake is between generative models, not between parameter settings within one. Historical work on the linguistic and institutional reorganisation of Lavoisian chemistry (Mcevoy, 1989; Perrin, 1988; Boantz, 2008) reinforces this: the revolution was a re-tooling of which questions could be posed, not a re-weighting of answers to fixed questions. A scalar order parameter cannot, by construction, carry this information, because the cardinality of admissible nodes and edges is exactly what is changing. Even philosophers sympathetic to a deflationary reading of incommensurability concede that the residue which resists deflation is structural rather than quantitative (Malone, 1993). The scalar treatment is therefore silent on what the historians take to be the substantive content of the revolution.

Failure modes the scalar treatment cannot diagnose. A second pressure comes from a different direction. Even if one set aside the question of structural reorganisation and asked only after the social pathologies a multi-agent paradigm model ought to recover, the scalar treatment runs into a distinction it cannot draw. Nguyen (2018) argues that two pathologies of collective inference – routinely conflated in popular usage – have entirely different mechanisms and admit entirely different remedies. An *epistemic bubble* is a failure of *informational topology*: relevant voices are omitted from the network, coverage is incomplete, and exposure to those voices is sufficient to dissolve the pathology. An *echo chamber* is a failure of *trust calibration*: outside voices are not absent but actively discredited, so that exposure to them, if anything, reinforces the chamber. The two failure modes can co-occur, but they are not the same object; follow-up work emphasises that they decouple in practice (Pollock, 2024) and that the trust-calibration variant can become self-stabilising in ways that omission alone cannot explain (Clinkenbeard, 2025). A scalar belief variable updated under socially-weighted observations registers both pathologies as the same kind of thing – a population whose marginal has collapsed onto one attractor – and so cannot, on its own resources, distinguish a community that has merely failed to encounter an alternative from one that has built the discrediting of alternatives into its updating rule. The two patients present with the same symptom, but they have different diseases, and the scalar model has no diagnostic on which to separate them.

Locating the gap. The Lavoisian case and the bubble/chamber distinction press from opposite ends on the same point. From the philosophy-of-science side, the object being reorganised in a paradigm shift is the inferential web, and a scalar carries none of the relevant structure. From the social-epistemology side, the pathologies a multi-agent treatment ought to distinguish are

differentiated by their structural locus – topology versus trust – and a scalar cannot mark which locus is failing. These are not arguments against the scalar treatment as such; they are arguments that something has to be added to it. What that something is, and how it can be added without giving up the dynamical traction the scalar treatment already buys, is the business of the next section.

5 Structural Extension: Bayes Nets and Bayesian Model Reduction

The scalar treatment of §3 sufficed to expose asymmetric belief-revision dynamics, but it cannot speak to what practising scientists actually argue about: which commitments are foundational, which are negotiable, and which channel of observation to spend a finite attentional budget on next. The structured extension recovers all three by promoting each agent’s paradigm from a point in Θ to a Bayesian network whose nodes carry theoretical commitments and whose edges encode inferential dependencies.

Structural belief representation. The conceptual ancestry of paradigms-as-networks predates active inference by half a century. [Wahyudi et al. \(2026\)](#), reconstructing Quine’s coherentism, summarises the web-of-belief picture in which no proposition confronts experience in isolation: revisions propagate along inferential links, and the choice of which link to cut is underdetermined by data alone. Lakatos’s methodology of scientific research programmes sharpens this into a topology with a designated *hard core* of commitments shielded by a flexible *protective belt* of auxiliary hypotheses ([Heijdra and Lowenberg, 1986](#); [Ostapiuk, 2025](#)); the belt absorbs anomalies so that the core remains in force, and a programme degenerates only when belt-patching ceases to predict novel facts. Thagard’s ECHO programme operationalises Quinean coherence as a connectionist constraint network and demonstrates, on the continental-drift case study, that the same machinery can reconstruct a historical paradigm shift as a re-equilibration of explanatory links ([Thagard, 1989](#); [Thagard and Nowak, 1988](#)); [Ranney and Thagard \(1988\)](#) extend the account to belief revision in naive physics. None of these accounts furnishes a generative model, but each picks out the same structural object: a graph of commitments whose densely-connected interior is held rigid while its periphery negotiates with observation. We adopt this object verbatim, identifying core nodes with the parameters under tight prior precision and belt nodes with those under broad prior precision in the agent’s generative model.

Bayesian model reduction. Promoting the core/belt distinction to a piece of formal machinery requires a principled treatment of *structural* inference — the problem of scoring alternative graphs, not just alternative parameter values, against the same data. [Friston et al. \(2018\)](#), extending the post-hoc model selection scheme of Friston & Penny (2011), provide precisely this in the form of Bayesian model reduction (BMR). The construction is straightforward in variational Bayes: once the full model has been inverted, the evidence and posterior of any *reduced* model — one differing only in its priors — can be computed analytically from the full posterior and the ratio of priors, generalising the Savage–Dickey density ratio. Concretely, switching off a candidate edge corresponds to collapsing the prior variance on the associated parameter to zero; the resulting change in variational free energy is the (log) Bayes factor for that structural revision. For the Dirichlet posteriors that govern categorical likelihood mappings in discrete state-space models — the natural form for theory–observation links — the reduced free energy has a closed form in terms of multivariate beta functions. [Friston et al. \(2018\)](#) showcase BMR as both a data-analytic tool and a metaphor for neurobiological abductive reasoning: the same equations that prune redundant DCM connections describe the synaptic elimination an agent performs when “thinking off-line” simplifies its model of a contingency. [Marković et al. \(2023\)](#) extend the

scheme to deep-network sparsification, evidencing that the reduction is general across parameter spaces. We invoke BMR as the engine of paradigm dynamics: candidate revisions to an agent’s network — adding a belt link, demoting a core node, accepting a rival auxiliary — are scored by their reduced free energy, with the structural affective utility inherited from §2 entering as a contribution to the cost of admitting each revision. Path-dependence in the scalar model thus refines into something graph-theoretic: revisions that touch the core are more expensive because more of the posterior has to be re-balanced.

Active sampling and precision allocation. The structural representation closes the loop with action selection in a way the scalar model could not. Observation in the structured model is not exogenous arrival of evidence but an active sampling decision: the agent allocates a finite precision budget across belt nodes, sampling some channels with high precision and others with low. [Friston et al. \(2015\)](#) establishes the underlying primitive — expected free energy decomposes into extrinsic (utility) and epistemic (information-gain) value — and [Millidge et al. \(2021\)](#) surveys the equivalent decompositions. [Mirza et al. \(2019\)](#) show, in an MDP formulation of Yarbus’s classic visual-search paradigm, that selective attention is just context-dependent modulation of the likelihood precisions ζ : raising ζ on task-relevant channels and lowering it on task-irrelevant ones reweights expected information gain so that the agent saccades to evidence that distinguishes hypotheses it actually entertains. The same construction underwrites [Feldman and Friston \(2010\)](#), [Parr and Friston \(2017\)](#), and the precision-control account of [Parr and Friston \(2019\)](#); [Limanowski and Friston \(2018\)](#) fills in the failure mode where mis-allocated precision dampens exploration. Transposed to the paradigm setting, precision allocation *is* the paradigm-dependence of observation: an agent inhabiting a Newtonian network attends to mass-and-force belt nodes with high precision and assigns negligible precision to channels whose only hidden cause lies in a rival network. The result is a structured analogue of theory-laden perception (§1) ([Hyland and Albarracín, 2025](#)), now derived rather than stipulated: what counts as a salient observation is what the agent’s current network has bandwidth to believe.

6 Derived Consequences of the Structured Model

The structural extension developed in Section 5 is not introduced for its own sake. It earns its place because three otherwise-disparate results in network epistemology — the Zollman tradeoff between consensus speed and asymptotic accuracy, the paradoxical-stubbornness finding from the division-of-cognitive-labor literature, and Nguyen’s distinction between epistemic bubbles and echo chambers — recover as derived consequences of one mechanism: a trust-precision network propagating candidate *structural* revisions rather than scalar opinions. We position each result in turn and indicate which observable in our simulations it indexes.

The Zollman tradeoff and the value of inefficiency. ?? showed, in a multi-armed-bandit network of Bayesian scientists, that *denser* communication graphs converge faster but more often lock onto the worse arm; sparser graphs sustain transient diversity long enough for the inferior hypothesis to be discarded, and so reach higher asymptotic accuracy. The mechanism is now well understood as a spectral one: the rate at which beliefs synchronise on a graph is set by its algebraic connectivity (the Fiedler value λ_2 of the graph Laplacian) ([Gripic, 2021](#); [Montijano et al., 2017](#); [Tan et al., 2008](#)), so larger λ_2 compresses the transient phase during which dissent can falsify a leading hypothesis. The structured model inherits this argument unchanged at the topological level, but reroutes its substantive content: the channels carry *precision-weighted candidate revisions* of the generative model, not scalar credences, so the speed–accuracy tradeoff applies to structural revision events themselves. This vindicates the Feyerabendian-pluralist intuition ([Müürsepp et al., 2019](#)) in mechanistic form — a measured inefficiency in the trust-precision network is the substrate condition for the critical-discursive

uptake that $\mathcal{P}$ requires of an epistemically healthy community — and supplies the formal claim Longino’s critical contextual empiricism has been criticised for leaving underspecified (Novaes and Çağlar Dede, 2025; Ajdari, 2023; Borgerson, 2011).

Division of cognitive labour and paradoxical stubbornness. $\mathcal{P}$ and $\mathcal{P}$ argued that the priority rule incentivises credit-driven scientists to spread across research programmes, producing a community-level distribution of effort closer to the social optimum than truth-driven scientists would individually choose. $\mathcal{P}$ relaxed the god’s-eye information assumption: in their local-information variant, agents that appear *paradoxically stubborn* — unwilling to abandon a low-success-probability programme — in fact regularise the system, preventing premature convergence on whichever programme happened to lead first. $\mathcal{P}$ qualifies the optimistic Kitcher–Strevens reading by noting that the priority rule rewards novelty regardless of replicability and so *disincentivises* the very direct-replication work that would discipline structural revisions. In the structured model these findings appear as a single statement: an agent’s resistance to a structural revision is proportional to the precision of its current generative model, so agents with strong structural utility on their incumbent paradigm act as protective elements against premature structural collapse. They are not uniquely irrational; their structural commitments simply require correspondingly stronger cross-channel evidence to revise. The Muldoon–Weisberg paradox and the Romero replication-vice fall out as two readings of the same asymmetry — one casting it as virtue (transient diversity preserved), the other as vice (replication suppressed) — and the model exposes the parameter that interpolates between them.

Bubbles versus chambers: failure modes of cross-agent channels. $\mathcal{P}$ draws a distinction that, in our framework, is structural rather than rhetorical. An *epistemic bubble* is a failure of *trust topology*: relevant voices are omitted, so the cross-agent channel is sparse or disconnected, and the bubble pops on simple exposure. An *echo chamber* is a failure of *trust precision*: the discrediting beliefs assigned to outsiders pre-emptively downweight incoming cross-surprisals, so the chamber survives exposure and can even use it for reinforcement (Pollock, 2024; Turner, 2023; Clinkenbeard, 2025). In the scalar-opinion setting this distinction must be argued for philosophically; in the structured-belief setting it is recovered as the formal separation between sparsity of $\mathbf{T}$ and the precision weights it carries on its non-zero entries. Crucially, because the channels now carry candidate *structural* revisions rather than scalar opinions, the bubble–chamber distinction becomes more honest: a bubble withholds candidate revisions; a chamber admits them and then assigns them zero precision. The two failure modes call for different repairs — coverage versus precision-recalibration — and the model makes the repair prescription mechanistic rather than virtue-theoretic.

7 Discussion

External attentional curation. The third open question is the most consequential for the model’s empirical domain of validity, and it remains open. Throughout Sections 5–6 an agent’s policy over which neighbours, papers, and observations to attend to is treated as endogenous: it is selected by expected free energy under the agent’s own generative model and affective utility. In contemporary scientific practice this assumption is already strained. Citation recommenders, search-engine ranking, social-media amplification, and journal-level visibility filters increasingly mediate which findings an individual scientist ever encounters, and the same logic operates a fortiori in adjacent epistemic communities — policy analysis, public discourse around scientific results, the training corpora of large models that themselves now participate in summarisation and triage. The literature on algorithmic curation documents that such systems do not merely accelerate access to a fixed information environment but actively reshape exposure through popularity bias, preference reinforcement, and the formation of “information cocoons” that insulate

users from disconfirming evidence (Lin, 2026; Zhao et al., 2020), and that the boundary between helpful personalisation and manipulation is structurally hard to draw once the recommender’s objective is decoupled from the user’s epistemic interest (D, 2025).

For the present construction the consequence is precise but unresolved. The attentional policy π no longer factorises through a single agent’s generative model: a curator $\mathcal{C}$ — algorithmic, institutional, or both — intervenes between the latent paradigm field and the per-agent observation channel, with its own objective that is in general not aligned with predictive accuracy on the underlying scientific question. The trust matrix $\mathbf{T}$ then learns precisions over a doubly-mediated signal, and the resource-flow readouts acquire a third upstream lever beyond the exogenous inflow rate and initial trust configuration: the curator’s policy itself. Whether this collapses to a renormalisation of the existing trust dynamics, or whether it introduces qualitatively new attractors — in particular, attractors that survive the corrective force of cross-surprisal because the disconfirming observations are never sampled — is the model’s natural next question. It is also the point at which the construction makes contact with broader concerns about AI systems gradually displacing human epistemic agency (Kulveit et al., 2025) and about the multi-principal coordination problems that arise when curators, scientists, and institutions optimise different objectives over a shared information substrate (Critch and Krueger, 2020). We flag this as the open direction the framework most naturally extends into, and the one we take to be most urgent.

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